

	PROJECT P5.0 Das Island		
	PIPING MATERIAL SPECIFICATION		
Job No. PE112	Doc. No. P5-100-A4L-8-08002-0001	Rev No. 0	PIPE CLASS: CE3L0C-JX

SYNOPSIS

Services:	PROCESS HYDROCARBON, PROCESS CONDENSATE, PRODUCED WATER, CHEMICAL		
Pipe Material:	LTCS clad	Flange Rating:	CL 600 RTJ
Valve Trim:	trim UNS N08825+stellite	PWHT:	THK. GREATER THAN 19MM
Corrosion Allowance:	0MM	Code:	ASME B31.3
Service Type:	sour	Special Requirements:	LETHAL (NOTE 48) + NACE MR0175 / ISO 15156 (NOTES 4, 13) & NOTE 148

SERVICE LIMITS

Size Range	0.75 - 24	0.75 - 24	0.75 - 24	0.75 - 24	0.75 - 24	0.75 - 24
Temperature (C)	-46	0	38	50	100	150
Pressure (BARG)	102.1	102.1	102.1	100.2	93.2	90.2
Size Range	0.75 - 24					
Temperature (C)	200					
Pressure (BARG)	87.6					

PRIMARY

Size	0.75	1	1.5	2	3	4	6	8
Sch/WT	Sch80S	Sch80S	Sch80S	Sch40S	Sch40S	Sch40S	Sch80/XS	Sch60
Size	10	12	14	16	18	20	24	
Sch/WT	Sch60/XS	Sch60	Sch60	Sch80	Sch60	Sch80	Sch80	

Item	Size Range	Rating	End Connection	Specifications	Notes
Pipe	0.75 - 4		BE	ASME B36.19M smls ASTM B423 UNS N08825 NACE MR0175/ISO 15156	1 , 85
Pipe	6 - 16		BE	ASME B36.10M smls ASTM A333 GR 6 clad UNS N08825 NACE MR0175/ISO 15156	
Pipe	18 - 24		BE	ASME B36.10M EFW(100% RT) ASTM A671 GR CC65 CL22 S2 clad UNS N08825 NACE MR0175/ISO 15156	
Elbow 45 deg	0.75 - 4		BW	ASME B16.9 LR smls ASTM B366 GR WPNICMC-S NACE MR0175/ISO 15156	2
Elbow 45 deg	6 - 16		BW	ASME B16.9 LR smls ASTM A420 GR WPL6 clad UNS N08825 NACE MR0175/ISO 15156	2
Elbow 45 deg	18 - 24		BW	ASME B16.9 LR welded(100% RT) ASTM A420 GR WPL6W clad UNS N08825 NACE MR0175/ISO 15156	2 , 82
Elbow 90 deg	0.75 - 4		BW	ASME B16.9 LR smls ASTM B366 GR WPNICMC-S NACE MR0175/ISO 15156	2
Elbow 90 deg	6 - 16		BW	ASME B16.9 LR smls ASTM A420 GR WPL6 clad UNS N08825 NACE MR0175/ISO 15156	2

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Elbow 90 deg	18 - 24		BW	ASME B16.9 LR welded(100% RT) ASTM A420 GR WPL6W clad UNS N08825 NACE MR0175/ISO 15156	2 , 82
Tee equal	0.75 - 4		BW	ASME B16.9 smls ASTM B366 GR WPNICMC-S NACE MR0175/ISO 15156	2
Tee equal	6 - 16		BW	ASME B16.9 smls ASTM A420 GR WPL6 clad UNS N08825 NACE MR0175/ISO 15156	2
Tee equal	18 - 24		BW	ASME B16.9 welded(100% RT) ASTM A420 GR WPL6W clad UNS N08825 NACE MR0175/ISO 15156	2 , 82
Tee reducing	0.75 - 10		BW	ASME B16.9 smls ASTM B366 GR WPNICMC-S NACE MR0175/ISO 15156	2
Tee reducing	8 - 16		BW	ASME B16.9 smls ASTM A420 GR WPL6 clad UNS N08825 NACE MR0175/ISO 15156	2
Tee reducing	18 - 24		BW	ASME B16.9 welded(100% RT) ASTM A420 GR WPL6W clad UNS N08825 NACE MR0175/ISO 15156	2 , 82
Reducer Concentric	0.75 - 10		BW	ASME B16.9 smls ASTM B366 GR WPNICMC-S NACE MR0175/ISO 15156	2
Reducer Concentric	6 - 16		BW	ASME B16.9 smls ASTM A420 GR WPL6 clad UNS N08825 NACE MR0175/ISO 15156	2
Reducer Concentric	18 - 24		BW	ASME B16.9 welded(100% RT) ASTM A420 GR WPL6W clad UNS N08825 NACE MR0175/ISO 15156	2 , 82
Reducer eccentric	0.75 - 10		BW	ASME B16.9 smls ASTM B366 GR WPNICMC-S NACE MR0175/ISO 15156	2
Reducer eccentric	6 - 16		BW	ASME B16.9 smls ASTM A420 GR WPL6 clad UNS N08825 NACE MR0175/ISO 15156	2
Reducer eccentric	18 - 24		BW	ASME B16.9 welded(100% RT) ASTM A420 GR WPL6W clad UNS N08825 NACE MR0175/ISO 15156	2 , 82
Cap	0.75 - 4		BW	ASME B16.9 smls ASTM B366 GR WPNICMC-S NACE MR0175/ISO 15156	2
Cap	6 - 24		BW	ASME B16.9 smls ASTM A420 GR WPL6 clad UNS N08825 NACE MR0175/ISO 15156	2
Nipoflange	4 - 24 * 1 - 1	CL 600	RTJ	mfg std ASTM B564 UNS N08825 150mm long NACE MR0175/ISO 15156	2 , 73
Nipoflange	4 - 24 * 2 - 2	CL 600	RTJ	mfg std ASTM B564 UNS N08825 150mm long NACE MR0175/ISO 15156	2 , 73
Nipoflange	4 - 24 * 1.5 - 1.5	CL 600	RTJ	mfg std ASTM B564 UNS N08825 150mm long NACE MR0175/ISO 15156	2 , 73
Weldolet	3 - 24 * 0.75 - 4			MSS SP-97 ASTM B564 UNS N08825 NACE MR0175/ISO 15156	
Weldolet	18 - 24 * 6 - 8			MSS SP-97 ASTM A350 GR LF2 CL1 clad UNS N08825 NACE MR0175/ISO 15156	
Flange WN	0.75 - 4	CL 600	WN RTJ	ASME B16.5 ASTM B564 UNS N08825 NACE MR0175/ISO 15156	2
Flange WN	6 - 24	CL 600	WN RTJ	ASME B16.5 ASTM A350 GR LF2 CL1 clad UNS N08825 NACE MR0175/ISO 15156	2

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Item	Size Range	Rating	End Connection	Specifications	Notes
Blind Flange	0.75 - 4	CL 600	RTJ	ASME B16.5 ASTM B564 UNS N08825 NACE MR0175/ISO 15156	
Blind Flange	6 - 24	CL 600	RTJ	ASME B16.5 ASTM A350 GR LF2 CL1 clad UNS N08825 NACE MR0175/ISO 15156	
Spectacle Blank	0.75 - 4	CL 600	RTJ	ASME B16.48 ASTM B424 UNS N08825 NACE MR0175/ISO 15156	3
Spectacle Blank	6 - 6	CL 600	RTJ	ASME B16.48 ASTM A516 GR 70 S5 clad UNS N08825 NACE MR0175/ISO 15156	3 , 189
Spacer & Blank	8 - 24	CL 600	RTJ	ASME B16.48 ASTM A516 GR 70 S5 clad UNS N08825 to suit ASME B16.5 NACE MR0175/ISO 15156	3 , 189
Insulating gasket kit	0.75 - 24	CL 600	RTJ	mfg std raised face (type F) GRE(G-11) laminated UNS N08825 core with GRE(G-11) sleeve with seal to suit ASME B16.5 NACE MR0175/ISO 15156	
Gasket Ring Joint	0.75 - 24	CL 600		type R octagonal Alloy 825 ASME B16.20 to suit ASME B16.5 NACE MR0175/ISO 15156	
Stud Bolt	0.75 - 6			ASTM A320 GR L7M/A194 GR 7M S3 with 2 heavy hex nuts dims to ASME B18.2.1/18.2.2 threaded to ASME B1.1 CL 2A/2B fluorocarbon polymer coated	
Stud Bolt	8 - 24			ASTM A320 GR L7M/A194 GR 7M S3 with 3 heavy hex nuts dims to ASME B18.2.1/18.2.2 threaded to ASME B1.1 CL 2A/2B fluorocarbon polymer coated	
Ball Valve	0.75 - 1.5	CL 600	full bore flg RTJ	Design to API 6D/ISO 14313 dims to API 6D/ASME B16.10 long pattern body ASTM B564 UNS N08825 floating ball UNS N08825+Tung carbide metal seat UNS N08825+Tung carbide side entry 2/3piece FE CL A lever NACE MR0175/ISO 15156	17
Ball Valve	2 - 4	CL 600	red bore flg RTJ	Design to API 6D/ISO 14313 dims to API 6D/ASME B16.10 long pattern body ASTM A494 GR CU5MCuC trunnion mounted ball UNS N08825+Tung carbide metal seat UNS N08825+Tung carbide side entry 2/3piece FE CL A lever NACE MR0175/ISO 15156	17
Ball Valve	2 - 4	CL 600	full bore flg RTJ	Design to API 6D/ISO 14313 dims to API 6D/ASME B16.10 long pattern body ASTM A494 GR CU5MCuC trunnion mounted ball UNS N08825+Tung carbide metal seat UNS N08825+Tung carbide side entry 2/3piece FE CL A lever NACE MR0175/ISO 15156	17
Ball Valve	6 - 6	CL 600	red bore flg RTJ	Design to API 6D/ISO 14313 dims to API 6D/ASME B16.10 long pattern body ASTM A352 GR LCC clad UNS N08825 trunnion mounted ball LTCS clad UNS N08825 metal seat UNS N08825+Tung carbide side entry 2/3piece FE CL A lever NACE MR0175/ISO 15156	17 , 23 , 21

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Item	Size Range	Rating	End Connection	Specifications	Notes
Ball Valve	6 - 24	CL 600	full bore flg RTJ	Design to API 6D/ISO 14313 dims to API 6D/ASME B16.10 long pattern body ASTM A352 GR LCC clad UNS N08825 trunnion mounted ball LTCS clad UNS N08825+Tung carbide metal seat UNS N08825+Tung carbide side entry 2/3piece FE CL A gear NACE MR0175/ISO 15156	17 , 23 , 21
Ball Valve	8 - 24	CL 600	red bore flg RTJ	Design to API 6D/ISO 14313 dims to API 6D/ASME B16.10 long pattern body ASTM A352 GR LCC clad UNS N08825 trunnion mounted ball LTCS clad UNS N08825+Tung carbide metal seat UNS N08825+Tung carbide side entry 2/3piece FE CL A gear NACE MR0175/ISO 15156	17 , 23 , 21
Gate Valve	0.75 - 1.5	CL 600	flg RTJ	Design to API 602/ISO 15761 std bore dims to ASME B16.10 bolted bonnet bellow sealed ASTM B564 UNS N08825 wedge gate trim UNS N08825+stellite FE CL A handwheel NACE MR0175/ISO 15156	17
Gate Valve	2 - 4	CL 600	flg RTJ	Design to API 603 std bore dims to ASME B16.10 bolted bonnet ASTM A494 GR CU5MCuC wedge gate trim UNS N08825+stellite FE CL A handwheel NACE MR0175/ISO 15156	17
Gate Valve	6 - 6	CL 600	flg RTJ	Design to API 600/ISO 10434 std bore dims to ASME B16.10 bolted bonnet ASTM A352 GR LCC clad UNS N08825 wedge gate trim UNS N08825+stellite FE CL A handwheel NACE MR0175/ISO 15156	17
Gate Valve	8 - 24	CL 600	flg RTJ	Design to API 600/ISO 10434 std bore dims to ASME B16.10 bolted bonnet ASTM A352 GR LCC clad UNS N08825 wedge gate trim UNS N08825+stellite FE CL A gear NACE MR0175/ISO 15156	17
Globe Valve	0.75 - 1.5	CL 600	flg RTJ	Design to API 602/ISO 15761 std bore dims to ASME B16.10 straight pattern body bolted bonnet bellow sealed ASTM B564 UNS N08825 trim UNS N08825+stellite FE CL A handwheel NACE MR0175/ISO 15156	17
Globe Valve	2 - 4	CL 600	flg RTJ	Design to API 623 std bore dims to ASME B16.10 straight pattern body bolted bonnet ASTM A494 GR CU5MCuC trim UNS N08825+stellite FE CL A handwheel NACE MR0175/ISO 15156	17
Globe Valve	6 - 12	CL 600	flg RTJ	Design to API 623 std bore dims to ASME B16.10 straight pattern body bolted bonnet ASTM A352 GR LCC clad UNS N08825 trim UNS N08825+stellite FE CL A gear NACE MR0175/ISO 15156	17
Check Valve	0.75 - 1.5	CL 600	piston flg	Design to BS 1868 dims to ASME B16.10 bolted cover body ASTM B564 UNS N08825 trim UNS N08825+stellite NACE MR0175/ISO 15156	

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Item	Size Range	Rating	End Connection	Specifications	Notes
Dual Plate Check Valve	2 - 4	CL 600	double flg RTJ	Design to API 594 retainerless dims to API 594 body ASTM A494 GR CU5MCuC trim UNS N08825+stellite NACE MR0175/ISO 15156	
Dual Plate Check Valve	6 - 24	CL 600	double flg RTJ	Design to API 594 retainerless dims to API 594 body ASTM A352 GR LCC clad UNS N08825 trim UNS N08825+stellite NACE MR0175/ISO 15156	
Monoblock DBB valve	0.75 - 1.5	CL 600	flg RTJ	Design to API 6D/ISO 14313 red bore floating ball type block needle os&y type bleed body ASTM B564 UNS N08825 ball UNS N08825+Tung carbide metal seat UNS N08825+Tung carbide FE CL B lever NACE MR0175/ISO 15156	90 , 306
Monoblock DBB valve	2 - 2	CL 600	flg RTJ	Design to API 6D/ISO 14313 red bore trunnion mounted ball type block needle os&y type bleed body ASTM B564 UNS N08825 ball UNS N08825+Tung carbide metal seat UNS N08825+Tung carbide FE CL B lever NACE MR0175/ISO 15156	90 , 306

Notes	
1:	IN LINE WITH P5-100-A4A-8-02915-0001 (PROCESS DESIGN BASIS) THE MINIMUM PIPING SIZE IS NPS 1. HOWEVER, THE LINE SIZE IN THIS CLASS NPS 3/4 & NPS 1/2 ARE INCLUDED FOR INSTRUMENT CONNECTIONS ONLY.
2:	ALL BUTT-WELDED COMPONENT THICKNESSES SHALL MATCH THE PIPE THICKNESS.
3:	FOR SPECTACLE BLINDS (FIG-8 FLANGES) & BLINDS REFER TO SPECIFICATION P5-100-A4L-8-028002-0001.
4:	ALL MATERIAL FOR SOUR SERVICE SHALL CONFORM TO THE REQUIREMENTS FOR MATERIAL SELECTION GUIDELINES P5- 100-A4Z- 8- 44400- 0001, REQUIREMENTS FOR MATERIALS IN SEVERE SERVICE P5- 100-A4Z- 8- 44402- 0001 AND NACE MR0175/ISO 15156 (FOR UPSTREAM) & NACE MR0103/ISO 17945 (FOR REFINERY SERVICE).
5:	PWHT SHALL BE BASED ON ASME B31.3 AND THE REQUIREMENTS OF SPECIFICATION P5-100-A4L-8-028002-0001 PIPING MATERIAL SPECIFICATION INDEX. FOR SOUR SERVICE PWHT REQUIREMENT SHALL BE ALSO GOVERNED BY HARDNESS CRITERIA AS PER NACE MR0175 / ISO 15156 (FOR UPSTREAM) & NACE MR0103/ISO 17945 (FOR REFINERY SERVICE).
13:	FOR SOUR/ LETHAL SERVICE 100%RT, 100%MT/PT HAS TO BE CONSIDERED IRRESPECTIVE OF RATING IN LINE WITH REQUIREMENT FOR MATERIALS IN SEVERE SERVICE P5- 100-A4Z- 8- 44402- 0001.
15:	EXTERNAL FASTENERS (BOLTS, STUDS & NUTS) SHALL BE COATED WITH FLUOROCARBON POLYMER SYSTEM AND SHALL COMPLY WITH SALT SPRAY TEST AS PER MATERIAL SELECTION GUIDELINE P5- 100-A4Z- 8- 44400- 0001.
17:	ALL VALVES IN LETHAL SERVICE SHALL MEET THE FUGITIVE EMISSIONS TEST AS PER BS EN ISO 15848 PART-1 & PART -2 WITH LEAKAGE CLASS 'AH' (REFER TO VALVE SPECIFICATION P5-100-A4L-8-08006-0001).
21:	IN PIPE CLASSES WITH DESIGN TEMPERATURE ABOVE -46 DEG. C, CLADDED VALVE OBTURATORS CAN BE ACCEPTED IN SIZE DN 100 AND ABOVE INSTEAD OF SOLID CRA.
22:	ALL PIPING COMPONENTS UP TO NPS 24 SHALL BE DESIGNED FOR VACUUM CONDITION AT AMBIENT TEMPERATURE. FOR HIGHER SIZES VACUUM DESIGN SHALL BE APPLICABLE IF INDICATED IN THE LINE LIST.
23:	WHEREVER ALLOY 825 TRIM IS SPECIFIED, COMPONENTS MADE FROM WELD OVERLAY SHALL USE INCONEL-625 WELDING CONSUMABLES. REFER AGES-SP-09-015.
43:	THREADED JOINTS ARE NOT PERMITTED.

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48:	PIPING DESIGN AND FABRICATION IN LETHAL SERVICE SHALL COMPLY WITH CHAPTER VIII (PIPING FOR CAT.M FLUID SERVICE) AND APPENDIX F OF THE PIPING CODE ASME B 31.3.
73:	NIPOFLANGE SHALL BE USED FOR THERMOWELL CONNECTION FOR HEADER NPS 4 AND ABOVE.
79:	PRESSURE-TEMPERATURE RATINGS FOR CLAD PIPE CLASSES SHALL SPECIFY LOWER PRESSURE TEMPERATURE RATING OF BASE PIPE MATERIAL OR SOLID CRA. BASE PIPING WALL THICKNESS SHALL BE VALIDATED OVER AND ABOVE SPECIFIED WALL THICKNESSES, IN CASE CLADDING / WELD OVERLAY IS NOT PRACTICAL BY APPROVED CLADDING / WELD OVERLAY VENDORS.
80:	THE PIPE THICKNESS ARE CALCULATED BASED ON P-T RATING TABLE FOR THIS CLASS, HOWEVER FOR SIZES NPS 26 AND ABOVE THICKNESS ARE CALCULATED BASED ON PROCESS DESIGN CONDITIONS.
81:	PIPING CLASS COVERS ALL TYPES OF VALVES NORMALLY USED IN THE INDUSTRY. HOWEVER, VALVE TYPE SELECTION SHALL BE AS PER PROCESS ISOLATION PHILOSOPHY (P5-100-A4A-8-02932-0001,P5-100-A4L-8-08006-0001) AND P&ID.
82:	WELDED PIPES AND WELDED FITTINGS SHALL BE 100% RADIOGRAPHED. WALL THICKNESS NEGATIVE TOLERANCES OF WELDED FITTINGS SHALL NOT BE LESS THAN WELDED PIPE.
84:	BOLT TENSIONING REQUIREMENT SHALL BE AS PER PIPING MATERIAL SPECIFICATION P5-100-A4L-8-028002-0001.
85:	SMALL BORE PIPE THE MINIMUM SCHEDULE SHALL BE AS PER P5-100-A4L-8-08000-0001 APPENDIX A1.
86:	DISSIMILAR FLANGE MATERIAL SHALL BE SEPERATED USING INSULATING GASKET, ONLY TO BE USED WHEN STATED IN CORROSION REPORT AND IN P&ID OR OTHERWISE WITH COMPANY APPROVAL. FOR HYDROCARBON SERVICE FIRE SAFE INSULATING GASKET IS MANDATORY (REFER P5-100-A4L-8-08010-0001 FOR INSULATING GASKET DETAILS).
90:	PIPING TO INSTRUMENT IDBB, FLANGED ON BOTH PROCESS SIDE AND INSTRUMENT SIDE.TO BE USED IN SOUR, TOXIC, AND VIBRATING SERVICE.
148:	PIPING CLASS CONTAINS COMBINATION OF SOLID CRA AND CARBON STEEL WITH CLAD MATERIAL. THE MINIMUM THICKNESS OF THE CLADDING (INCLUDING WELD OVERLAY) SHALL BE 3.0MM WITH A TOLERANCE OF +2.0MM, -0.0MM. REFER AGES-SP-09-015 FOR CLADDING REQUIREMENT.
189:	CRA CLAD BLANKS IS ACCEPTABLE INSTEAD OF SOLID CRA FOR SIZE NPS 6 AND ABOVE.
306:	DBB VALVES IN SOUR OR TOXIC OR HYDROCARBON OR LETHAL SERVICE SHALL MEET FUGITIVE EMISSION TESTING REQUIREMENTS AS PER BS EN ISO 15848 PART-1 & PART-2 WITH LEAKAGE CLASS 'BH' (REFER TO VALVE SPECIFICATION P5-100-A4L-8-08006-0001).

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SYNOPSIS

Services:	PROCESS HYDROCARBON, PROCESS CONDENSATE, PRODUCED WATER, CHEMICAL		
Pipe Material:	LTCS clad	Flange Rating:	CL 600 RTJ
Valve Trim:	trim UNS N08825+stellite	PWHT:	THK. GREATER THAN 19MM
Corrosion Allowance:	0MM	Code:	ASME B31.3
Service Type:	sour	Special Requirements:	LETHAL (NOTE 48) + NACE MR0175 / ISO 15156 (NOTES 4, 13) & NOTE 148

BRANCH TABLE

90° BRANCH CONNECTIONS

ABBREVIATIONS:

- E : TEE_WROUGHT-EQUAL
- T : TEE_WROUGHT-REDUCING
- W : OLET-WELDOLET

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BRANCH PIPE SIZE

RUN PIPE SIZE		0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	24
	24	W	W	W	W	W	W	W	W	T	T	T	T	T	T	E
	20	W	W	W	W	W	W	W	T	T	T	T	T	T	E	
	18	W	W	W	W	W	W	W	T	T	T	T	T	E		
	16	W	W	W	W	W	W	T	T	T	T	T	E			
	14	W	W	W	W	W	W	T	T	T	T	E				
	12	W	W	W	W	W	W	T	T	T	E					
	10	W	W	W	W	W	T	T	T	E						
	8	W	W	W	W	W	T	T	E							
	6	W	W	W	W	T	T	E								
	4	W	W	T	T	T	E									
	3	W	W	T	T	E										
	2	T	T	T	E											
	1.5	T	T	E												
	1	T	E													
	0.75	E														